

# SANTOSH KUMAR KASHYAP

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## Education

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| <b>Malaviya National Institute of Technology, Jaipur</b><br><i>M.Tech in Computer Science and Engineering</i> | <b>Aug 2022 – Jun 2024</b><br><i>CGPA: 8.47/10</i> |
| <b>Himachal Pradesh Technical University, Hamirpur</b><br><i>B.Tech in Computer Science and Engineering</i>   | <b>Aug 2017 – Jun 2021</b><br><i>CGPA: 8.99/10</i> |

## Experience

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| <b>Tecorb Technologies</b><br><i>AI/ML Engineer</i> | <b>Nov 2024 – Present</b><br><i>Noida, Uttar Pradesh</i> |
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- Intelligent Adaptive Tutoring System :** Architected a dual-source parallel RAG pipeline using MongoDB Atlas HNSW Vector Search and `all-MiniLM-L6-v2` (384-d), reducing retrieval latency by **50%** ( `500ms` → `250ms`) through concurrent retrieval across agent-specific and shared institutional knowledge.
- Engineered a deterministic cosine-similarity relevance gate ( $\geq 0.3$ ) that blocks LLM generation when retrieved context falls below the relevance threshold, **eliminating LLM token usage for out-of-scope queries** and reducing unsupported generation.
- Designed an adaptive RL-inspired query-routing policy across a 4-action space using  **$\epsilon$ -greedy exploration** ( $\epsilon = 0.2$ ) and softmax action selection over student confusion states, with a **DPO-inspired tabular feedback update loop** bounded to 2 iterations.
- Built a 2-tier caching subsystem with thread-safe L1 in-memory TTL caching and L2 Redis across 4 Uvicorn workers, achieving **1ms response time for cache-hit queries**, alongside a secondary Llama 3.1 8B evaluator auditing responses across 5 quality dimensions.
- Brain Renting :** Architected an async 5-stage cognitive distillation pipeline in FastAPI transforming unstructured expert corpora into 12-dimensional reasoning blueprints, achieving **100% PII redaction** across 40 planted canaries and a sub-0.0024 Membership Inference score (verbatim overlap  $\leq 12$  words).
- Engineered an empirical LLM evaluation harness benchmarking persona fidelity across 10 personas, demonstrating **49/50 blind identification accuracy**, **100% inter-persona separability** (30/30 pairs), 1.0x MMLU capability retention, and 1.10x IFEval instruction-following retention.
- Implemented multi-layered LLM safety and jailbreak guardrails via prompt hierarchy decoupling, achieving a **100% refusal rate** across 198 HarmBench vectors and eliminating autobiographical hallucinations (20/20 honest disclosures).
- Built a containerized multi-channel agent runtime (**cogent-engine**) streaming inference across WhatsApp, Telegram, and Slack via LiteLLM dynamic routing, incorporating advisory-register gating (**97.5% compliance**) and 6-turn context retention ( $\geq 0.85$ ).
- Developed production marketplace backend on PostgreSQL 17 and Docker Compose, integrating Stripe Connect double-sided token wallets, field-level AES-256-GCM encryption with HMAC blind indexing, and native 3D AST knowledge graph synthesis.

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| <b>Shorthills AI</b><br><i>Software Development Engineer (SDE 1)</i> | <b>Sep 2024 – Aug 2025</b><br><i>Gurugram, Haryana</i> |
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- Best Views Reviews (BVR):** Developed LLM-powered conversational chatbots for an AI-driven product research platform analyzing 18.6M+ product reviews across 1,500+ categories, enabling users to interact with product and review insights conversationally.
- Built RAG-based conversational workflows using LangChain, Elasticsearch, and LLMs, retrieving relevant product/review information and generating context-aware responses.
- Implemented prompt engineering, tool/function calling, conversation memory, product recommendation workflows, and streaming responses to build interactive chatbot experiences.
- PartsGenie:** Contributed to automated PDF ingestion and preprocessing pipelines, including OCR, text extraction, and metadata parsing to prepare unstructured documents for RAG workflows.
- Developed a RAG-based chatbot using LangChain and LLMs to answer natural-language queries over multi-hundred-page automotive manuals and technical documents.
- Integrated Elasticsearch-based retrieval with LLM workflows and implemented prompt engineering, tool/function calling, and conversational context for grounded responses.
- Developed an AI-powered Root Cause Analysis (RCA) chatbot to surface likely failure causes and relevant historical fixes from manuals and maintenance data.

- Worked on ML and NLP projects involving text processing and classification using Python and machine learning techniques.

## Research Publication

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### SafeCode: Enhancing Data Race Detection in OpenMP Through Hyperparameter-Optimised DCNN [\[IEEE\]](#)

- Developed a deep learning framework for detecting data races in OpenMP programs using Abstract Syntax Trees (ASTs), text vectorization, embeddings, and a Deep Convolutional Neural Network (DCNN).
- Applied random-search hyperparameter optimization with Early Stopping, Model Checkpointing, and ReduceLROnPlateau to improve model performance.
- Achieved 89.60% accuracy and 89.60% F1-score on synchronization-based datasets, outperforming the baseline CNN approach.

## Projects

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### Sentiment Analysis & MLOps | *Python, PyTorch, ONNX, FastAPI, Docker* | [GitHub Link](#) | [Demo Link](#)

- \* Fine-tuned RoBERTa-base on 120K customer reviews using Cosine Annealing, Stochastic Weight Averaging (SWA), and FP16 mixed precision, achieving 84.8% accuracy and 84.6% weighted F1 on held-out data.
- \* Engineered post-training decision calibration to prioritize negative-class detection, increasing Negative Recall from 79.1% to 90.34% and Negative F1 to 81.16%.
- \* Quantized the transformer to INT8 ONNX, reducing RAM usage by 58% and model size by 75%, while achieving 25–35 ms P50 CPU inference latency; deployed via asynchronous FastAPI with ONNX → PyTorch → GBDT fallback.
- \* Implemented real-time MLOps drift monitoring using sliding-window Kolmogorov–Smirnov and Chi-square tests to detect distribution shifts and trigger automated alerts.

### Fact Verification Engine | *Python, Spacy, Litellm, PyTorch, FastAPI* | [GitHub Link](#) | [Demo Link](#)

- \* Architected a real-time RAG pipeline that retrieves evidence from Wikipedia and DuckDuckGo, providing citation-backed fact verification without relying on stale vector databases.
- \* Engineered hybrid lexical and semantic re-ranking using BM25Okapi with phrase/exact-match boosting and PyTorch Cross-Encoder fallback, achieving sub-millisecond BM25 re-ranking.
- \* Built an atomic claim-extraction engine using SpaCy dependency parsing and integrated multi-provider LLM inference through LiteLLM and Groq, with structured JSON validation and few-shot verification guardrails.
- \* Deployed an asynchronous FastAPI service with Docker, two-tier LRU caching, TTL-based persistence, and JSONL telemetry for stage-level latency, cache-hit, and verification analytics.

## Technical Skills

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**Programming Languages:** Python

**AI/ML & GenAI:** LLMs, RAG, LangChain, LiteLLM, Prompt Engineering, Function Calling, Conversational AI, MCP

**ML Libraries/Frameworks:** PyTorch, TensorFlow, scikit-learn, MLflow

**Databases & Search:** SQL, MySQL, FAISS, Weaviate, Elasticsearch

**Backend:** Django, FastAPI

**Cloud & Tools:** AWS, Docker, Git

## Achievements

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**Gold Medal:** B.Tech (Computer Science & Engineering), Himachal Pradesh Technical University, 2021.

**Competitive Programming:** Solved 900+ algorithmic problems on LeetCode and GeeksforGeeks.

**GATE CS 2022:** Qualified Graduate Aptitude Test in Engineering (Computer Science).